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CubeSats detumbling using only embedded asymmetric magnetorquers

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Abstract

After deployment from a rocket, a CubeSat is detumbled using magnetorquer rods bringing the norm to the point where the reaction wheels take over to reduce the angular velocity to null. Therefore, utilizing reaction wheels for satellite detumbling at higher initial velocities is vital but they are heavy and occupy significant space on a spacecraft having challenging control. To address this challenge, this paper features a disruptive approach that conducts the control only by the PCB-integrated magnetorquers with various geometries using a diverse non-unity track width ratio. The trace widths are parametrized such that the optimal torque to power dissipation ratio is investigated. The optimizations are then simulated for various geometric distributions and validated through comprehensive measurement setups that establish a framework for selecting the best-case coil configuration according to mission requirements. The detumbling rates of multiple asymmetric coil configurations are compared with the embedded designs in published literature and state of the art. It is found that the proposed asymmetric embedded magnetorquers can detumble the vehicle at high initial angular velocities. Lastly, the simulation results of thermal analysis are validated for selecting the application-specific optimal coils configuration. At the end, the proposed system is compared with the embedded magnetorquers available in the literature and commercial attitude control systems.

Keywords: Magnetorquers; Air-Coil; Detumbling; Planar coils; Thermal Analysis

1. Introduction

A CubeSat is a miniaturized version of satellite frequently built due to low cost and high modularity, typically used to gather geographic readings, atmospheric radiation and navigation & communication. These small satellites are composed of cube units (U) of 10 cm × 10 cm × 10 cm in the form of 1U, 2U, 3U or even 12U. These miniaturized satellites were not originally designed to set standards. The concept originated in 1999 as a design challenge for graduate students of California Polytechnic University and Stanford University as a part of their research; however, currently this program has 21 years of satellite technological experience. The goal is to serve as a platform for university students to learn and develop the next-level miniaturized spacecraft that are capable of being reused for subsequent missions. European space agency (ESA) and the National Aeronautics and Space Administration (NASA) are also collaborating and assisting various small satellites into orbits. "The Space Economy Report 2020", Euroconsult guesses that the combined space economy, including both government space investments, as well as commercial space, totaled US$ 385 billion in 2020, a record amount, and is expected to double by 2030. There is no generalized qualification standard as the mechanisms of CubeSats are mainly custom made designed for a single mission (Kramer and Cracknell, 2008). Consequently, universities working on small satellites give possibilities to multi-satellite missions which were otherwise impossible

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because of their adaptability, modularity and scalability. The readily accessible high-performance commercial off the shelf (COTS) devices have revolutionized the research and development activities of the academic institutions and small & medium enterprises (SMEs) in the field of small satellites (Ahmed Khan et al., 2021b). Specifically, universities around the globe have a strong focus on the development of small satellites and have initiated many projects in last decade (Ahmed khan et al., 2022). At present, around 377 small satellites are in queue to be launched according to the Satellite Launch Demand Database (LDDB) (Cuadrado, 2017).

The major attraction of miniaturized spacecraft is having faster developmental times and low-cost projects; however, the reduced lifespan and higher failure probability (roughly 40 % for university satellites) are acknowledged disadvantages associated with these projects (Monkell et al., 2018). Thus, the attitude determination and control system (ADCS) also is subject to this constraint and is one of the prominent sources of failure. Therefore, refining the reliability of the ACS for a vehicle considerably mitigates the failure rate and increases the success ratio (Byeon et al., 2019; Li et al., 2017; Zhang et al., 2016). The size and mass of ADCS are mostly defined by the actuators and sensors (Ali et al., 2018b, 2018a). It is thus imperative to miniaturize these electromechanical systems without sacrificing attitude maneuverability. The block-level illustration of printed magnetorquer module's intercommunication with OBC subsystem is subcategorized in Fig. 1. Attitude determination systems determine the attitude by computing the difference between the real and desired attitude via attitude control algorithms utilizing magnetometers, sun trackers and gyroscopes. The commands via the controllers activate the actuators for generating the intended control torque to reduce the real and desired attitude difference e.g., reaction wheels, propulsion thrusters and magnetorquers, etc.

The stabilization and maneuverability of the spacecraft's attitude are classified into passive stabilization and active stabilization. Passive stabilization relies on the action of external torques, permanent magnets, solar radiation pressure, and gravity gradients. Passive control systems are often simple, requiring no power or moving parts but suffer from the drawback of poor pointing accuracy (Byeon et al., 2019; Zhang et al., 2016). Active stabilization uses magnetorquers, thrusters and reaction wheels for the orientation of the satellite. These systems depend on the complex dynamics of control laws with limited onboard processing, but allow the satellite to be more precisely controlled if needed for three-dimensional maneuverability.

The recent successful launch of LituanicaSAT-2 built in collaboration between Nanoavionics and Vilnius University demonstrated the chemical propulsion system's efficacy for nanosatellites (Ahmed khan et al., 2022). However, these chemical propulsion systems face issues of propellant toxicity, leakage and dependence on exothermic combustion, which affect the viability and usabiliy of the system. The field of electric micropropulsion is expanding at a tremendous rate in the form of miniaturized prototype versions of existing sophisticated designs (Wright and Ferrer, 2015). Multitude of novel micropropulsion technologies such as ion thrusters, Hall thrusters, pulsed plasma thrusters, colloid thrusters, laser ablation thrusters and helicon plasma thrusters are being developed for miniaturization under the stringent CubeSat standards (Levchenko et al., 2018). The endurance testing, low power management and standard miniaturization of pulsed plasma thrusters (PPT) are detailed in (Kvell et al., 2014). The robustness and reliability of the hall type thrusters in combination with permanent magnets are investigated in (Lorello et al., 2018). Irrespective of the ongoing research trend in the electric propulsion field, similar challenges to chemical propulsion limit the potential of these systems such as; reliability, robustness, non hazardous interaction with the spacecraft, radiation hardness, development speed, energy efficiencies and adaption to highly miniaturized CubeSat standard electric propulsion systems. Although, thrusters provide precise attitude maneuvers but given the small size of a CubeSat, most universities mainly use COTS based magnetorquers with a combination of reaction wheels for dumping the initial angular velocities. When the spacecraft is initially deployed after launch, magnetorquer rods are usually used to remove the initial tumbling rate to the level where the reaction wheels are turned ON and magnetorquers then are excited again to desaturate the reaction wheels (Giulietti et al., 2006). Decent magnetic moments

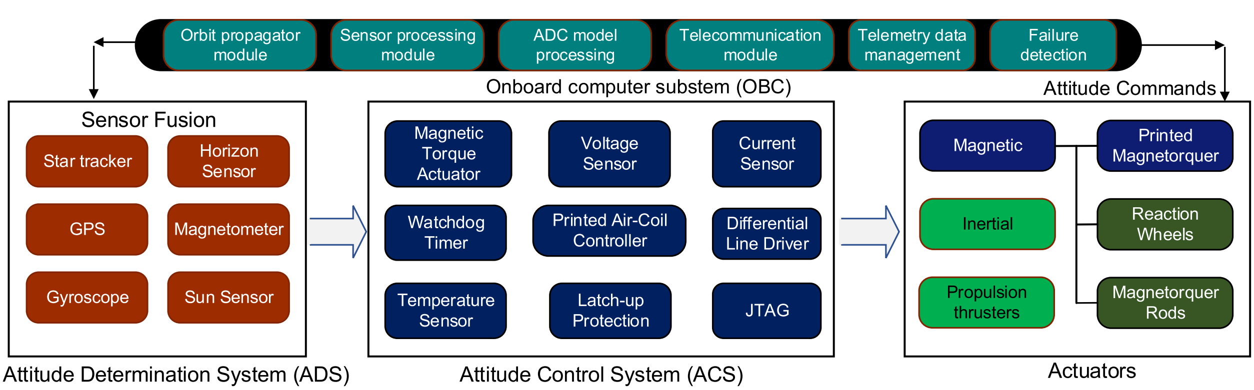


Fig. 1. Block level illustration of ADS and ACS intercommunication with onboard computer system (OBC).

are generated by these COTS based actuators; nevertheless, they are heavy and occupy significant space on a vehicle. In attitude and orbital control systems (AOCS), 50 % percent of faults occur because of the moving parts like that in thrusters, reaction wheels and gyroscopes (Tafazoli, 2009). Moreover, the reliability of lubrication systems (Jin et al., 2012) to support moving parts and their fault detection, isolation and identification (FDII) schemes to accurately diagnose the root cause of the fault in a complex non-linear system is a challenging task in small satellites (Rahimi et al., 2020).

Anwar et al and mughal et al used a disruptive method to replace the traditional 3D magnetic torquer rods into pseudo-2D symmetric embedded planar magnetorquer coils in the form of printed circuit board (PCB) copper traces (Ali et al., 2015; Mughal et al., 2020). These planar magnetorquers are almost massless because they are integrated inside the internal layers of PCB and offer better torque versus power dissipation ratio (Ahmed Khan et al., 2021a). However, due to their inherent restricted actuation torques, the challenge of designing fast initial detumbling maneuvers for a CubeSat is non-trivial. Driven by this constraint, it is crucial to investigate the actuation capabilities of embedded magnetorquers for dumping of high initial angular rates after spacecraft deployment. In this work, the projection based form of the widely adopted and standard B-dot controller reported in numerous sources will be used for the CubeSat detumbling (Lovera, 2015).

The embedded magnetorquers in the form of asymmetric planar windings to date have not yet been investigated for small satellites (Ali et al., 2014). This paper proposes novel PCB-integrated asymmetric winding geometric distributions as shown in Fig. 2 for the spacecraft detumbling maneuvers.

2. Mathematical model

2.1. Attitude equations of motion

The general parameterizations of attitude can be represented by the three coordinates or parameters. Therefore, the Euler angles as such are a minimal set of three attitude parameters. However, the Euler angles can give the same orientations in many ways that lead to singularities. As such, the choice of unit quaternion for attitude parameterization is motivated by its nonsingular representation, free of rotational sequences and offers computational simplicity. Moreover, they are less expensive computationally than the direction cosine matrix (Junkins and Schaub, 2009). The quaternion are then represented as a scalar (q_0) and a three element vector (q_1, q_2, q_3). The kinematics based upon the quaternion gives the orientation in space and is given through angular velocity's numerical integration. Hence, quaternion kinematics can be represented using quaternions as shown in (1).

$$\begin{Bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \begin{Bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{Bmatrix} \tag{1}$$

Where q_i shows the four quaternions and p, q, r represents the components of angular velocity in the body frame of satellite. The quaternions will be utilized to derive the transformation matrix from the inertial to body frame as inertial frame is under the focus. The angular velocity's derivative is computed by equating the accumulated sum of magnetic moments imparted to the spacecraft.

$$\begin{Bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{Bmatrix} = [I]^{-1} \left(\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} - \boldsymbol{S}(\vec{\omega}_{B/I})[I] \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \right) \tag{2}$$

The skew symmetric operator is denoted by $\mathbf{S}()$ and its multiplication with a vector gives a cross product.

$$\mathbf{S}(\vec{\omega}_{B/I}) = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \tag{3}$$

The moment of inertia is represented by I and is taken along in the body frame's centre of mass which is positive and symmetric. The magnetic moments are represented by the L, M and N generated by the three magnetorquers

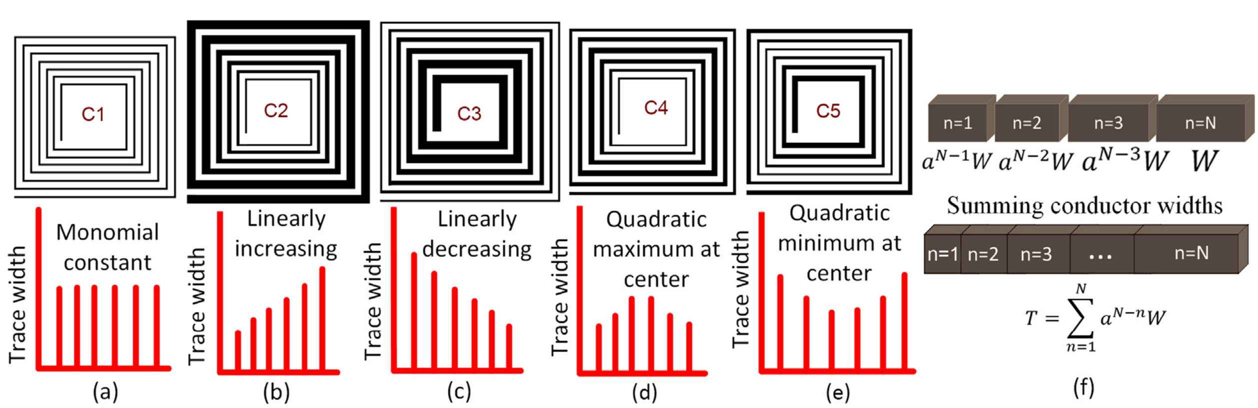


Fig. 2. (a)-(e). Magnetorquer windings layouts with five arrangements based on mathematical functions. Fig. 2 (f). TWR applied to trace widths s cross section, where T is the total width of traces in one layer which is identical for a given footprint and number of turns.

which offer a control torque to rotate the satellite as presented in (2). To avoid computational complexity, the torques in (2) have contributions only from the coils. Parasitic torques in the form of residual magnetic dipoles, solar radiation, aerodynamic and gravity gradient torques are neglected.

2.2. Asymmetric magnetorquer model

The spacecraft excited by embedded magnetorquers is required to be maneuvered by at least three coils, placed mutually orthogonal to each other in such a way that each coil is aligned with x-, y-, z-axis respectively. The winding distributions are integrated into the internal PCB layers. Magnetic moment is generated when the current is applied in the coils such that $\overrightarrow{i}_B = \left[i_x, i_y, i_z\right]^T$ and each coil can be controlled independently. The magnetic dipole generated by the coils as a result of applied current interacts with the geomagnetic field resulting in a control torque. The dipole magnetic moment $\overrightarrow{\mu_B}$ is then given below;

$$\overrightarrow{\mu_B} = nA\overrightarrow{i}_B \tag{4}$$

where each magnetorquer's number of turns is represented by n and A is the encompassed area.

The generated torque $\overrightarrow{\tau_B}$ in body reference frame by all the coils is given as the cross product of Earth's magnetic field with the magnetic dipole moment.

$$\overrightarrow{\tau_B} = \mathbf{S}(\overrightarrow{\mu_B})\mathbf{T}(\overrightarrow{q})\overrightarrow{B}_I \tag{5}$$

The quaternion transformation matrix $\mathbf{T}(\overrightarrow{q})$ is used to obtain spacecraft's vector of magnetic field from the inertial frame in the body frame as shown below;

$$T(\overrightarrow{q}) = \begin{bmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_1q_2+q_0q_3) & 2(q_1q_3-q_0q_2) \\ 2(q_1q_2-q_0q_3) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_1q_2+q_2q_3) \\ 2(q_0q_2+q_1q_3) & 2(q_2q_3-q_0q_1) & q_0^2-q_1^2-q_2^2+q_3^2 \end{bmatrix} \tag{6}$$

Eq. (5) can be rewritten utilizing the identity property following $\overrightarrow{a} \times \overrightarrow{b} = -\overrightarrow{b} \times \overrightarrow{a}$.

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = nA\begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix}\begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} \tag{7}$$

Where in the spacecraft's body frame, the components of magnetic field are represented by B_X, B_y, B_z.

Magnetorquer coils based on various winding geometric distributions in the form of metal traces are embedded in the multi-layered PCB designed with varying track width ratios. The magnetorquers' layouts with the asymmetric windings according to the mathematical functions are depicted in Fig. 2. The copper metal widths of Coils; C1, $C2$ and C3 are designed based on the monomial constant, linearly increasing and linearly decreasing functions, respectively. The Coils C4 and C5 follow the quadratic polynomial having the metal width being maximum and minimum in the central turns. Furthermore, a reconfigurable circuit based on MOSFETS switches is connected to the coil actuator for making the design reconfigurable (Mughal et al., 2020). The magnetic dipole moments and power dissipation in case of reconfigurable performance parameters are computed using the following equations.

The magnetic dipole moment $\overrightarrow{D}$ of M magnetorquer coils configured in any desired arrangement (n-series, m-parallel, n × m-hybrid) is given by.

$$\overrightarrow{D} = M\overrightarrow{D_0} \tag{8}$$

Where $\overrightarrow{D_0}$ represents magnetic moment of a single coil. Similarly, single coil's power consumption is then given by.P_0

$$P_0 = I_0^2 R_0 \because R_0 = \rho\frac{L_t}{A}$$

Power dissipation of M coils arranged in any desired arrangement is (n-series, m-parallel,n × m-hybrid) is;

$$P = MP_0 \tag{9}$$

$I = mI_0,\ V = nV_0,.M = n \times m$

The ratio between magnetic dipole moment generated and power dissipated is represented by the following equations.

$$\frac{D_0}{P_0} = \frac{N.S.I_0}{R_0.I_0^2} = \left(\frac{N.S}{R_0}\right).\frac{I}{I_0} \tag{10}$$

$$D_0^2 = \tfrac{(N.S)^2}{R_0}.P_0$$
$$D^2 = n.m.\tfrac{(N.S)^2}{R_0}.P = c.P \tag{11}$$

Where c or $\frac{D^2}{P_0}$ is a performance evaluation parameter that depends on the number of coils and their dimensions connected in series or a parallel configuration. As the subsequent number of coils is increased, the power dissipation and magnetic dipole moment increase whether in parallel, series or hybrid configuration. The current passing through the coils in case of single or series remains the same with similar generated torque.

2.3. B-Dot controller

A projection based method is applied to find the toque projection in the range space of the actuators for comparing the detumbling rates of various coil configurations which relies on the magnetorquer to use the simple control law (Monkell et al., 2018). This control law works by the imitation of vicious friction of spacecraft angular rotation in relation to the geomagnetic field B vector. The control law is given as following;

$$\overrightarrow{\mu_B} = K\mathbf{S}(\overrightarrow{\omega}_{B/I})\mathbf{T}(\overrightarrow{q})\overrightarrow{B}_I \tag{12}$$

The scalar control denoted by K is the gain factor. Torque is effectively exerted in the direction orthogonal to both the magnetic field and angular velocity which reduces the kinetic energy of the tumbling spacecraft. The expression depicts that the varying Earth magnetic field times a constant gain can damp the tumbling rates per orbit of the vehicle. By using (4), the current can be expressed in the component form using the identity following $\overrightarrow{a} \times \overrightarrow{b} = -\overrightarrow{b} \times \overrightarrow{a}$ (Monkell et al., 2018).

$$\begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} = \frac{K}{nA}\begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix}\begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \tag{13}$$

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = -K\begin{bmatrix} B_y^2+B_z^2 & -B_xB_y & -B_xB_z \\ -B_xB_y & B_x^2+B_z^2 & -B_YB_z \\ -B_xB_z & -B_yB_Z & B_x^2+B_y^2 \end{bmatrix}\begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \tag{14}$$

The aim is to reduce the rotation rates of the vehicle in detumbling state to zero. The control torque is not generated if the magnetorquers' axes are aligned with the local Earth's magnetic field and usually the reaction wheels are utilized to produce this component of control torque. However, the magnetic field of the Earth is time varying and keeps changing due to the satellite orientations around all the axes during detumbling state. Furthermore, coils are integrated in the internal layers of the PCB on which the solar panels are mounted on the top and the coil driver on the bottom, thus not taking up any extra space. So, only three solar panels PCBs are needed for three-dimensional maneuverability if they are mounted along the ×, y and z axes, without relying on the bulky and costly reaction wheels. Thus, it is clear that the B-Dot control law is able to effectively stabilize the spacecraft's angular momentum provided that the angular velocity $\overrightarrow{\omega}_B$ is not parallel to $\overrightarrow{B}_I$.

3. Printed magnetorquer coil driver

The magnetorquer module has been designed primarily with the low-cost COTS components readily available in the market. The COTS-based design enables academic and private institutions to improve the developmental practices that provide the testbed and experimental setup. The coil driver unit for the asymmetric magnetorquers comprises of STM32 microprocessor, housekeeping sensors (voltage, current and temperature), voltage regulators, latch-up protection system, Joint Test Action Group (JTAG) debugger, D-Sub connectors, differential line drivers MOSFETs based reconfigurable circuit and magnetic actuator for coil excitation. The magnetorquer module PCB is fabricated for the experimental verification of reconfigurability, current drawn, resistance and thermal dissipation of the various winding distributions.

The magnetorquer copper traces are integrated in the inner layers of multilayer solar panel PCB which does not occupy extra space as depicted in Fig. 3(a). Fig. 3(b) shows the magnetorquer driver PCB for actuation of Coils. The coil winding distributions having a varying number of turns takes the maximum available space on the tile (0.9 × 0.9 cm). The coils shown in Fig. 4 are experimentally excited for a range of 1–3.3 V inputs modeled after the CubeSats power distribution bus. The magnetorquers' magnetic actuation is controlled through the Dual H bridge motor driver IC (DRV8848, n.d.) by controlling the direction and amount of current flow as a function of applied voltage as shown in Fig. 5. The device is space qualified for low Earth orbit (LEO) which fully features the over temperature, short circuit overcurrent protection (OCP) and undervoltage-lockout (UVLO). The current is sustained through the gate drive in the MOSFETs by analog overcurrent protection called the deglitch time. The nFAULT pin is driven low when the analog current limit takes longer than the deglitch time of the OCP and all the MOSFETS are turned off. The IC will stay in OFF state till the retry time takes place, thus providing ultra-low quiescent current draw. The nFAULT pin drives low when the die temperature exceeds operational thermal limits and the MOSFETS are turned off. The operation is resumed normally when the nFAULT pin is released after the temperature falls to a safe level Similarly, when the VM pin's voltage is lower than the threshold voltage of the UVLO, the IC's circuity is turned off by resetting the internal logic. The coil actuation begins after VM rises above the UVLO rising threshold. The coil actuator 'DRV8848′ IC is connected to the designed reconfigurable circuit enabling the vehicle with a modular and reconfigurable ACS.

A unipolar, high-Side current measurement IC (INA168, n.d.) is employed to regulate the current as a function of differential input voltage, sending the data to the onboard computer through telemetries. Low-cost passive housekeeping sensors (temperature and voltage) are installed to manage the printed magnetorquer's power consumption and temperature rise under the operational thermal envelope.

An external watchdog (Max6373, n.d.) timer is employed that supervises the "STM32" microcontroller memory-access based activity and triggers a signal to restart the tile's central microprocessor if the system malfunctions. JTAG (PicoBlade, n.d.) test access port is incorporated in the system design that enables the consumer for debugging and testing of the ACS circuitry to ensure robust operation.

The mathematic model of the Earth's magnetic field is modeled using the International Geomagnetic Reference Field (IGRF). Sun tracking is modeled by following montenbruck and Plfeger's method (Montenbruck and Pfleger, 2000). The controller transmits signals depending upon the torque for desired rotation and maneuverability. The closed loop feedback sustains the required attitude control by reiterating the torque instruction until the intended stability and orientation are achieved. The required orientation and stability are achieved by the

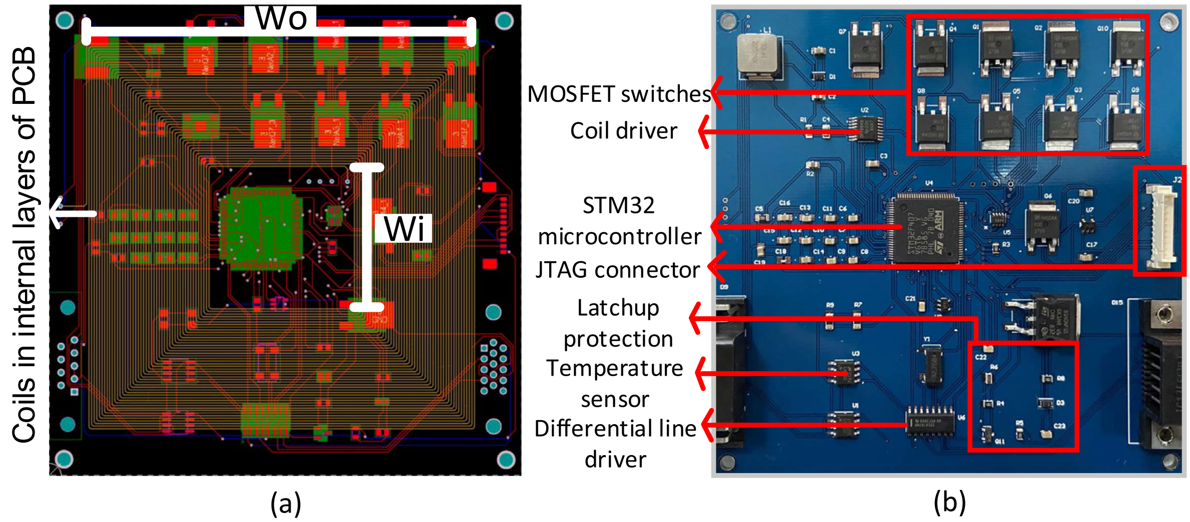


Fig. 3. (a) Magnetorquer copper traces embedded inside the internal layers of PCB. Fig. 3 (b) Magnetorquer coil driver PCB prototype.

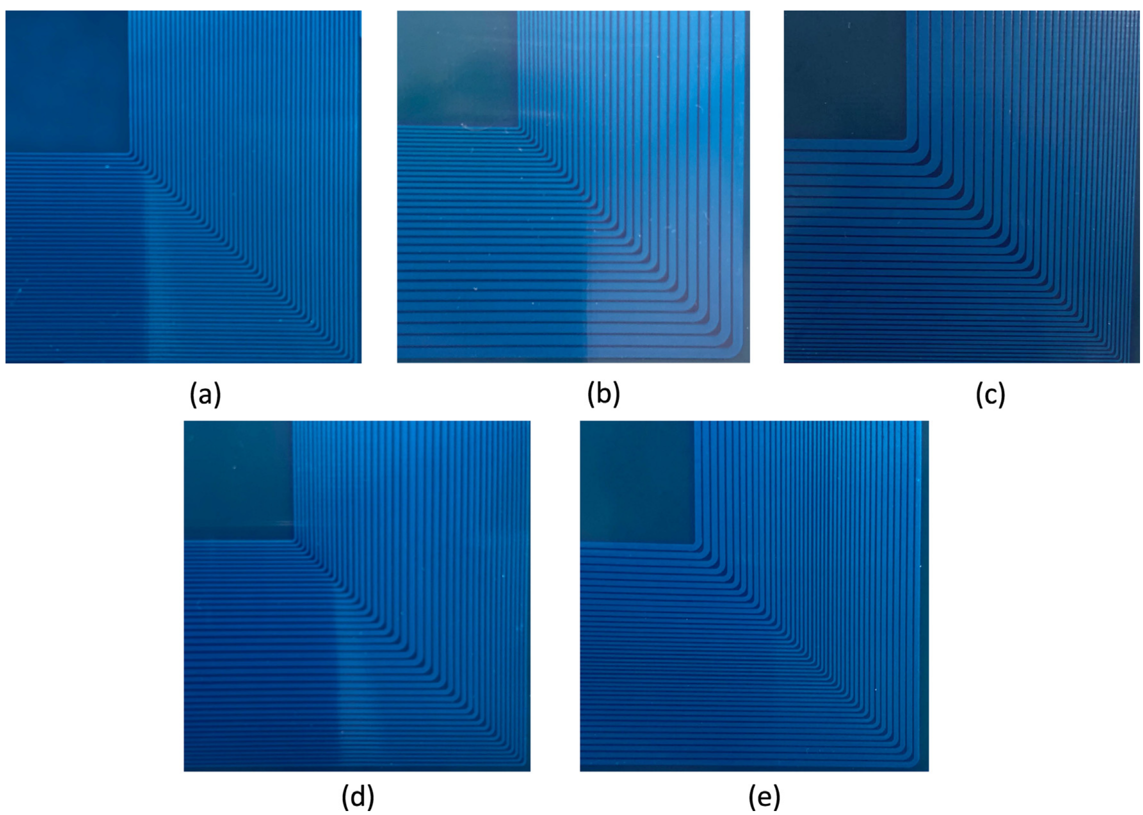


Fig. 4. Fabricated planar magnetorquer PCBs with TWR a = 0.95 applied to; (a) monomial constant C1 (b) linearly increasing $C2$ (c) linearly decreasing C3 (d) quadratic maximum C4 and (e) quadratic minimum C5 at central turns.

reiteration of the torque instructions through the closed loop feedback.

A latch-up is an instantaneous momentary effect in which the devices undergo into short-circuit state and high current passes from the power supply to the ground in the components, causing permanent damage to the system. Latch-ups in space occur more frequently owing its fact to the solar radiations making the COTS microdevices which are CMOS based vulnerable to damage. Incorporating bipolar microdevices can rectify this design flaw up to some extent as high energy is required to trigger the latch-up event. However, the microprocessors which are CMOS based still dominate the overall module design, so a latch-up proof design is simulated and developed according to (Ali et al., 2018b) making the designed coil driver module radiation hardened.

The printed magnetorquer module unit intercommunicates with the OBC subsystem of spacecraft through a dual

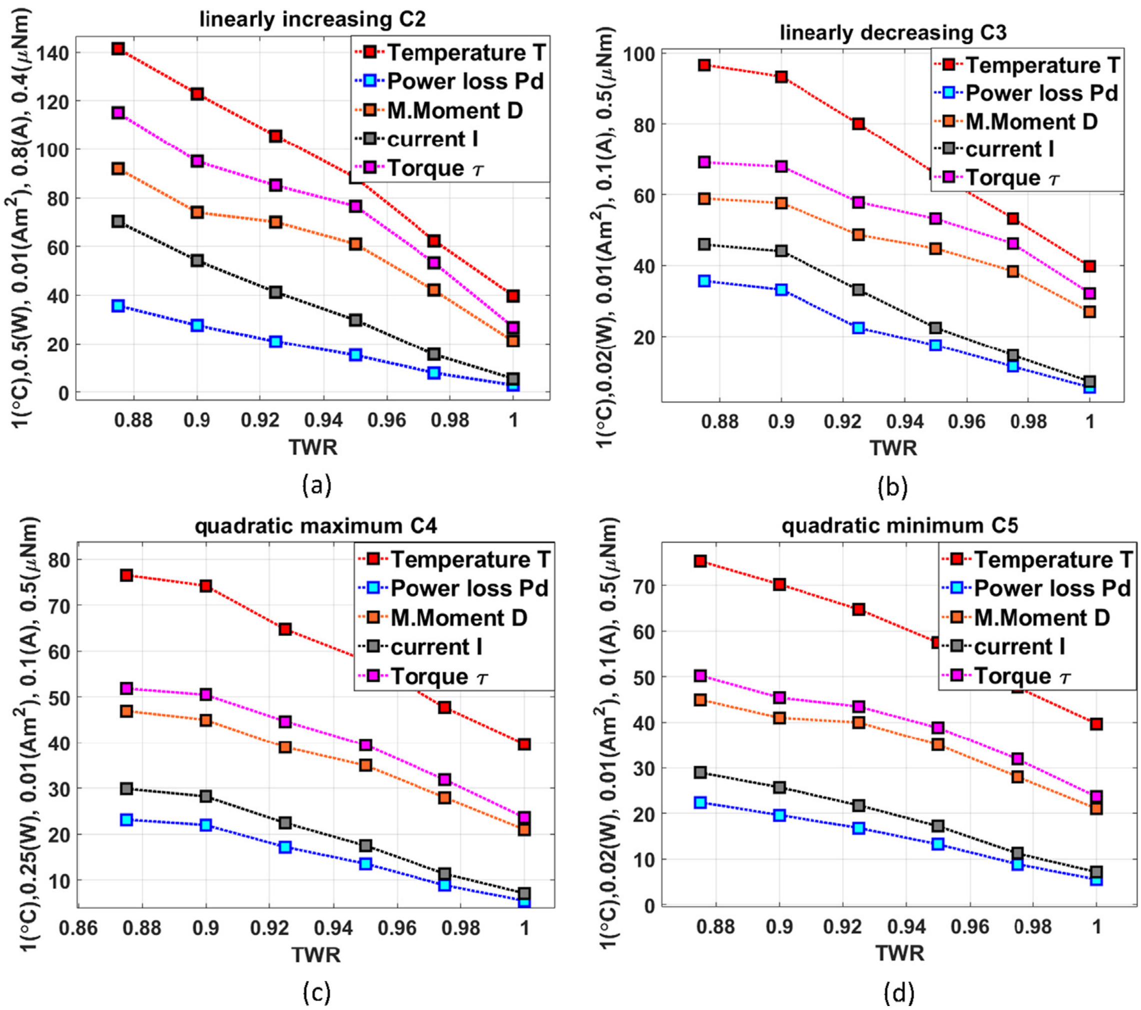


Fig. 5. TWR variation against key performance parameters for (a) linearly increasing *C*2 (b) linearly decreasing C3 (c) quadratic maximum C4 and (d) quadratic minimum C5 at central turns.

redundant CAN interface by using differential line drivers for all the telemetries and telecommands connected with D-Sub connectors. The coil driver and rest of the coprocessors are provided with sufficient telemetries (on the redundant CAN bus protocol) to ensure robust power management and operation of the system. High hysteresis and high input impedance are made possible because of the optimized differential line drivers which improve the noise immunity for accurate device control and precise bus communication, operating at switching frequencies up to 32 MHz, providing low power consumption without sacrificing the speed.

4. Effect of trace width variation

In this section asymmetric planar magnetorquer coils with various geometric distributions are simulated for testing the efficacy of developed layouts to design optimal coil configurations for a satellite with the primary goal of maximizing the torque versus power dissipation ratio. Hence, five planar magnetorquer coils with varying widths are simulated for testing the efficiency of the coils for a satellite. Equal maximum available space on the PCB tile (0.9 × 0.9 cm) is occupied by all the coils while the remaining space (0.1 cm) is left for the bypass diodes and temperature sensors. In this study, magnetic flux density, surface losses, currents and thermal envelope of multiple single layer PCB integrated asymmetric coil structures are analyzed using the AC/DC module of the COMSOL multiphysics. As the copper trace width is set to be 0.3 mm with the thickness of 5.4×10^{-9}, the coils are modeled as a boundary instead of a part with volume utilizing the Electric Currents in Shells interface. For evaluating the magnetic field distribution and surface density in the volume around the magnetorquer coil, the magnetic fields interface with the electric current in shells interface is used for analyzing the necessary performance parameteres. The width of the copper trace is varied according to the layouts based on mathematical functions (monomial constant, linearly increasing, linearly decreasing and quadractic maximum and minimum), while the space between the metal widths is maintained identical as illustrated in the layouts of Fig. 2. For a fair comparison of the performance characteristics, the inner and outer dimensions *wi* and *wo* for the non-uniform geometries are at equal distance and maintained identical. The winding resistance for the square and circular windings with variable metal width is calculated through the methods presented in (Cove and Ordonez, 2015; Huang et al., 1995).

The range of trace width ratio (TWR) values from 0.875 to unity with a difference of 0.25 is applied to more than twenty non-uniform coils for the evaluation of key performance parameters. When the TWR values approach unity, magnetic moment and torque decrease with the subsequent decrease in power dissipation and temperature. The current flowing through the trace widths as a function of applied voltage is responsible for the imparted torque to the spacecraft. As evident from Fig. 5, the sharp reduction of resistance slows down as the applied scaling factor or TWR values approach unity, which is an insightful observation. The authors used a multilayer PCB for embedding the magnetorquers to maximize the dipole moments and torques (Ali et al., 2015; Mughal et al., 2020). On the other hand, by using the proposed asymmetric magnetorquers, higher torques can be generated from only a single layer of PCB. The performance parameters of $C2$ and C3 are slightly different while the C3 and C4 almost show a similar behaviour regarding the resistance, current drawn, torque generated and power dissipation. Single layer non-uniform geometries key parameters are tabulated in Table 1.

Scaling factor $a = 0.95$ is applied for designing metal widths to selected coil arrangements ($C2$, C3, C4, C5) for evaluating the surface losses and magnetic flux density, as illustrated in Figs. 6 and 7. When the trace widths are made narrower, the magnetic field density distribution quickly increases over those spots, resulting in higher surface density losses. The magnetic field density distribution increases over the spots where the trace widths are decreased, resulting in increased resistance and higher surface density losses. Similarly, the resistance is decreased with the increase in trace widths resulting in higher current to flow, which generates higher magnetic moments but dissipates more power as well. It is thus evident that a tradeoff exists between surface losses and flux density distributions of the coil. For limiting the PCB design hotspots to maintain the quality factor, it is thus recommended to use the TWR ratio closer to unity for designing geometries with higher turns and narrow metal widths, resulting in a lesser effect of current crowding.

The magnetorquer based on an ideal geometric structure depends upon the tradeoff of efficiency, precise attitude maneuverability, power consumption and satellite dimensions, making it completely application-specific. Thus, the required power management of the mission should decide the selection of TWR values. To conclude, vehicles that need exceptional pointing accuracy with faster detumbling rates should design the magnetorquers with lower TWR values ($a = 1 \rightarrow 0$) i.e., farther from unity. Moreover, magnetorquers with narrow metal widths designed with TWR values closer to unity have greater inductance, less heat and thus better magnetic energy storage capabilities, suited for missions of longer spans. The non-uniform magnetorquers' single layer design performance parameters are tabulated in Table 1.

5. Simulations and results

Two CubeSats are simulated in this section for testing the efficacy of the proposed asymmetric actuators. The satellites are modeled according to $10 \times 10 \times 20\text{cm}^3$ (2U) and $10 \times 10 \times 30\text{cm}^3$ (3U) CubeSat dimensions with a mass of 2.8 and 5 kg. The inertia is computed to be $I_X = 0.0047\&I_Y = I_z = 0.011\text{kgm}^2$ and $I_X = 0.0083\&I_Y = I_z = 0.0417\text{kgm}^2$ for 2U and 3U CubeSats, respectively. The gain of the controller is calculated based on the design parameters of the various coils using (12). The simulations are carried out for a satellite placed in a 54° inclination orbit at an altitude of 400 km. As a result of separation from the launch vehicle, the initial detumbling rate is set to [1.2, −1.2, 1.2] (rad/s) which is about $68^\circ/sec$ for all the magnetorquers. Fixed step Runge-Kutta-4 integration routine is followed for simulation with a time-step of 0.1. Three solar panel PCBs mounted on the outside periphery of the satellite with the magnetorquers embedded in the four internal layers are considered with the assumption that current can be sent in both directions. In this way, three-dimensional attitude maneuverability is achieved. It is important to note

Table 1
Parameters of single layer printed asymmetric magnetorquers.

Values at $a = 0.95$	Parameters				
Coils	Monomial Constant, C1	Linearly increasing, $C2$	Linearly decreasing, C3	Quadratic maximum C4	Quadratic minimum C5
Inner dimensions, Wi, mm	12.5	12.5	12.5	12.5	12.5
Outer dimensions, Wo, mm	90	90	90	90	90
Single coil resistance, Ω	18.8	8.6	9.3	13	13.06
Surface electric potential, V	1.64	0.83	2.46	1.52	1.63
Total magnetic energy, μJ	14.93	63.62	32.52	16.39	27.6
Surface loss density, $\frac{W}{m^2}$	326.54	1298.1	799.17	441.74	625.59
Magnetic flux density norm, mT	0.81	1.55	1.22	0.93	1.14
Number of turns, N	63	33	33	43	43

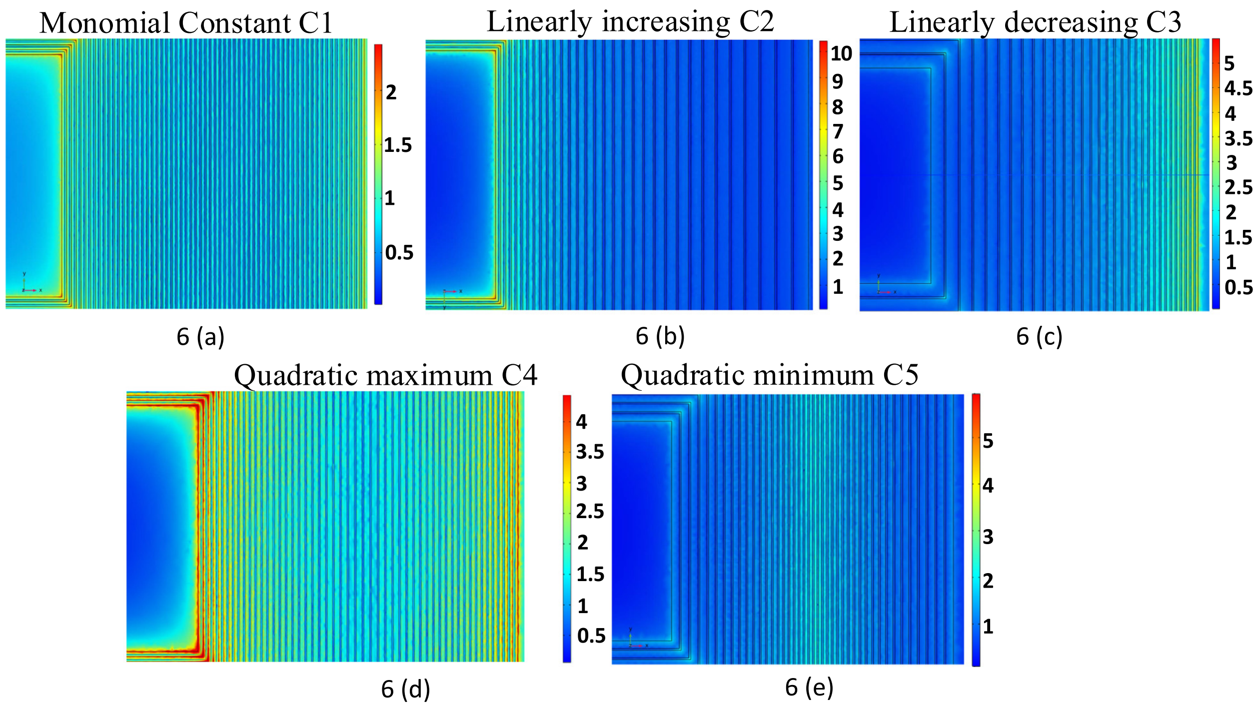


Fig. 6. Plots of Magnetic flux norm (mT) (above) and Surface loss density (below) with TWR a = 0.95 applied for (a) monomial constant C1 (b) linearly increasing *C*2 (c) linearly decreasing C3 (d) quadratic maximum C4 and (e) quadratic minimum C5 at central turns.

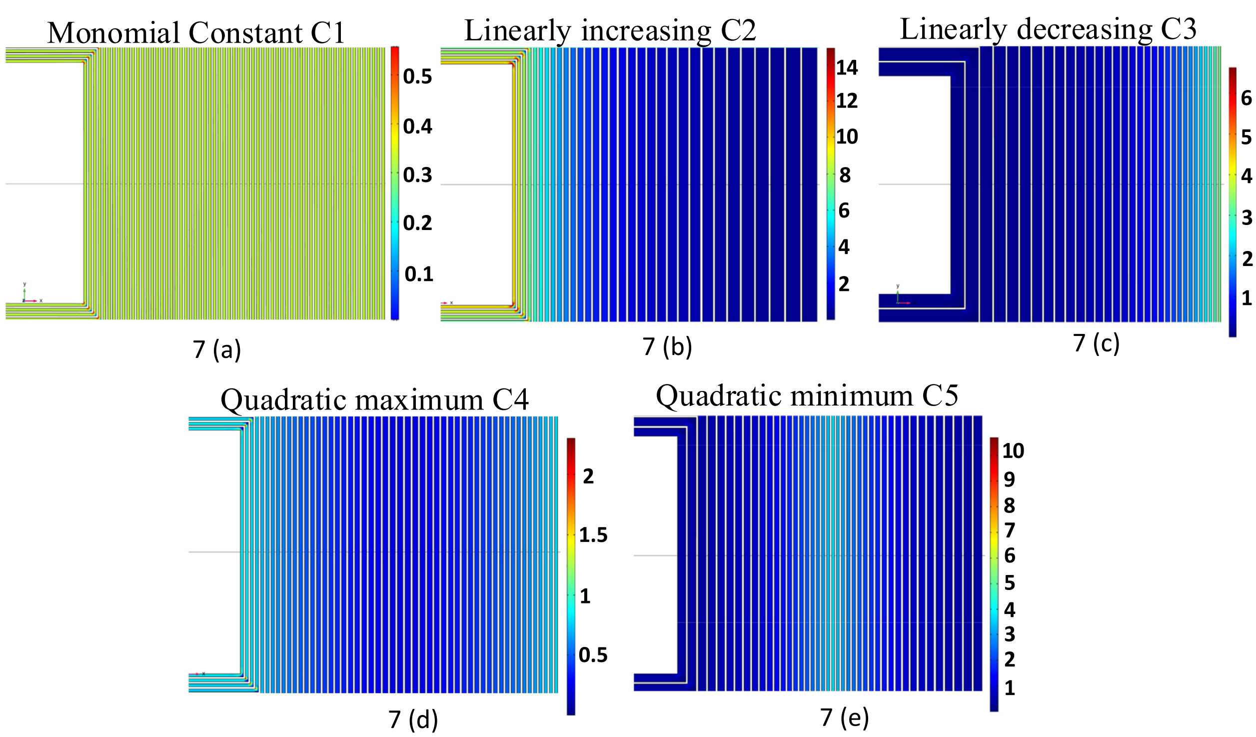
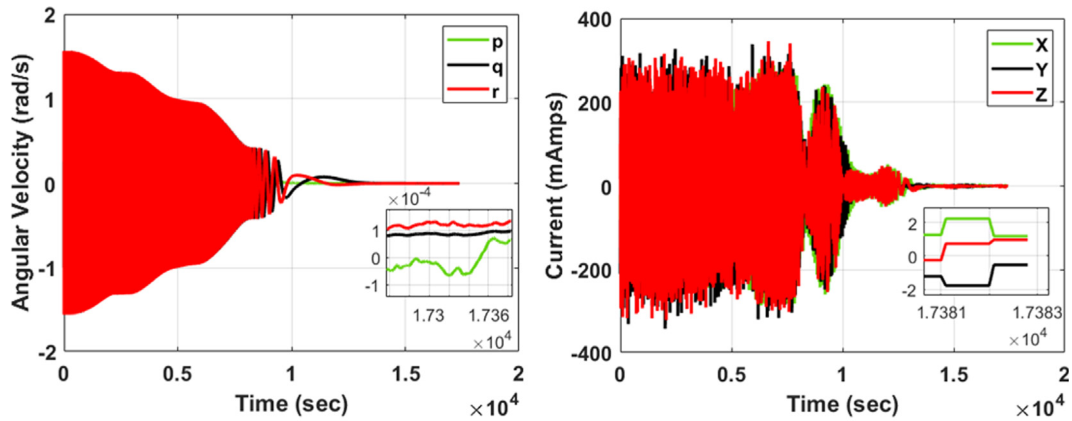


Fig. 7. Plots of Magnetic flux norm (mT) (above) and Surface loss density (below) with TWR a = 0.95 applied for (a) monomial constant C1 (b) linearly increasing *C*2 (c) linearly decreasing C3 (d) quadratic maximum C4 and (e) quadratic minimum C5 at central turns.

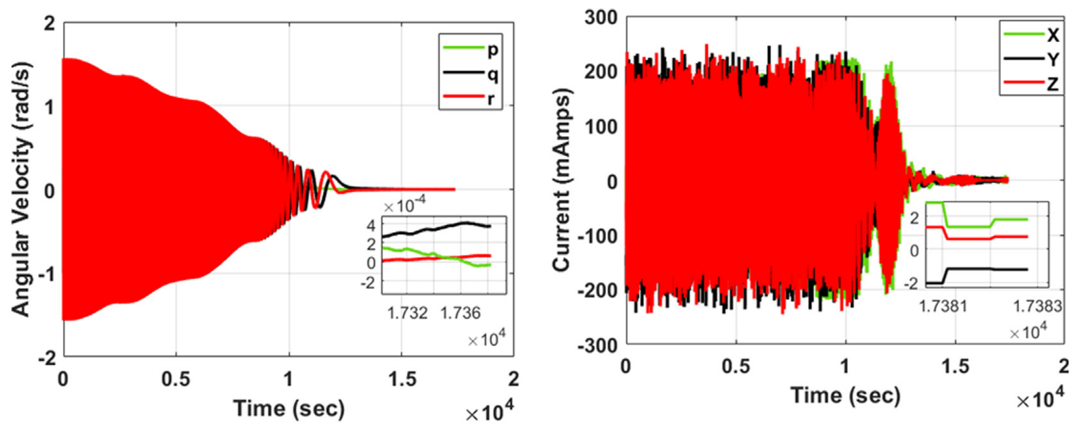
that if the angular velocity vector or the magnetic moment vector are collinear, the resulting current supplied to the magnetorquers will be zero according to (13) and consequently, the torque applied to the satellite is zero according to (12). However, this is usually not the case since the geomagnetic field is time and spatially varying which changes over time due to spacecraft's tumbling over all axes as well as changes in the spacecraft orbit.

Figs. 8-14 highlight key factors about the performance of coils for 2U and 3U CubeSats. All the proposed asymmetric coils detumbles the satellite with convergence rate based on the number of turns and the generated torque as a function of the flowing current. The greater the generated torque, the faster the detumbling rate. The linearly increasing coil $C2$ in parallel configuration (1×4) generates the highest torque and consequently detumbles the



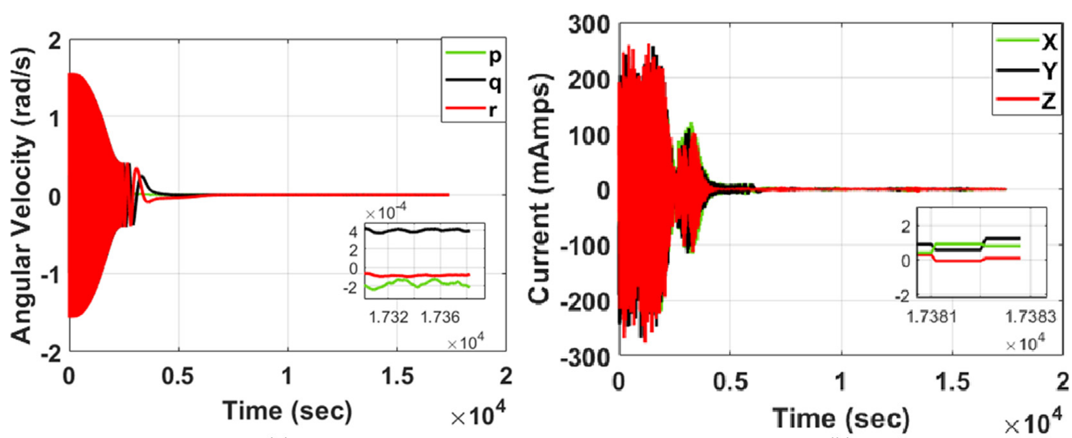
(a) (b)

Fig. 8. (a) 2U CubeSat s pitch, yaw and roll rate (rad/s) vs time (s) for linearly increasing C2. Fig. (b) 2U CubeSat s current drawn in pitch, yaw and roll axes for linearly Increasing C2.



(a) (b)

Fig. 9. (a) 2U CubeSat s Pitch, yaw and roll rate (rad/s) vs time (s) for quadratic maximum C5. Fig. 9 (b) 2U CubeSats current drawn in pitch, yaw and roll axes for quadratic maximum C5.



(a) (b)

Fig. 10. (a) 2U CubeSat s pitch, yaw and roll rate (rad/s) vs time (s) for linearly increasing C2 (1x4). Fig. 10 (b) 2U CubeSat s current drawn in pitch, yaw and roll axes for linearly increasing C2 (1x4).

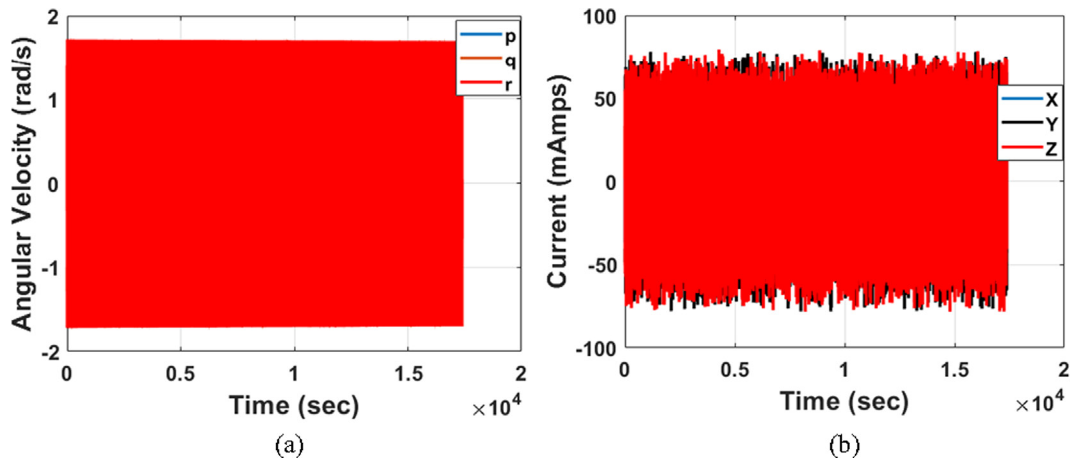


Fig. 11. (a) 2U CubeSat s pitch, yaw and roll rate (rad/s) vs time (s) for monomial constant C1. Fig. 11 (b) Current drawn in pitch, yaw and roll axes for monomial constant C1.

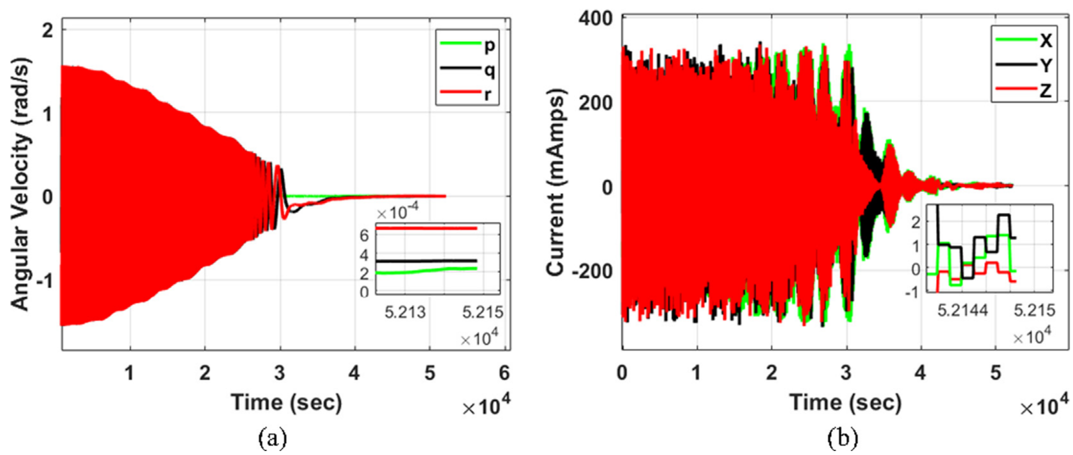


Fig. 12. (a) 3U CubeSat s pitch, yaw and roll rate (rad/s) vs time (s) for linearly increasing C2. Fig. 12 (b) 3U Cubesat s current drawn in pitch, yaw and roll axes for linearly Increasing C2.

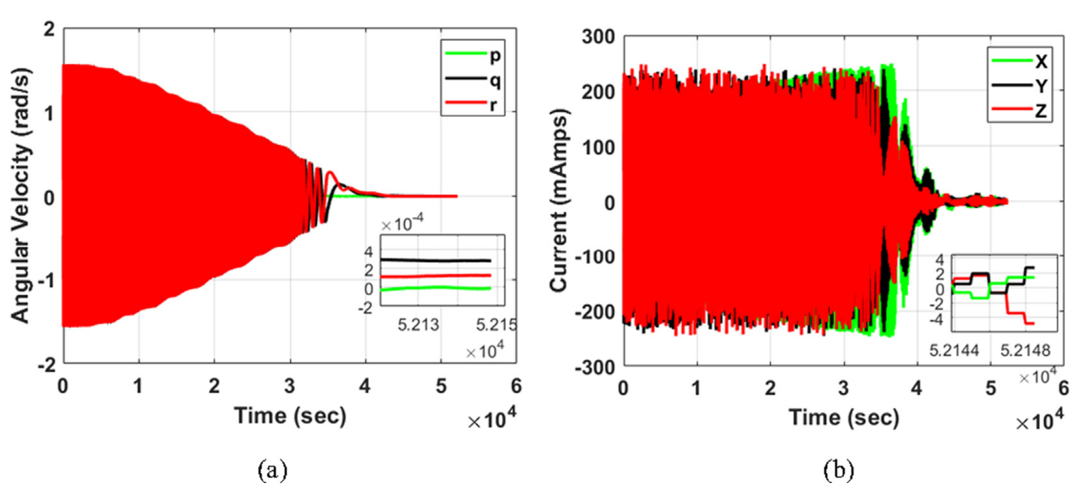


Fig. 13. (a) 3U Cubesat s pitch, yaw and roll rate (rad/s) vs time (s) for quadratic maximum C5. Fig. 13 (b) 3U Cubesat s current drawn in pitch, yaw and roll axes for quadratic maximum C5.

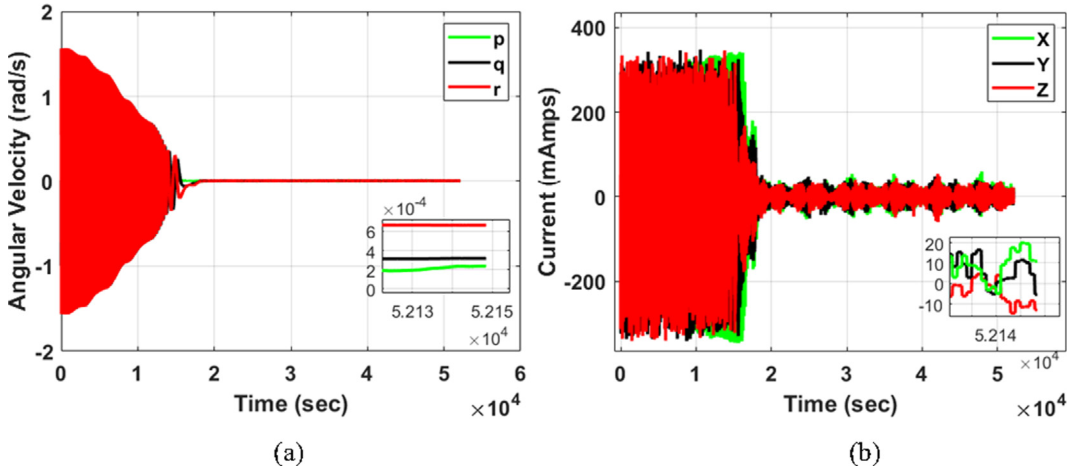


Fig. 14. (a) 3U CubeSat s pitch, yaw and roll rate (rad/s) vs time (s) for linearly increasing C2 (1x4). Fig. 14 (b) 3U CubeSat s current drawn in pitch, yaw and roll axes for linearly increasing C2 (1x4).

spacecraft faster but consumes more power and increases the temperature profile, as shown in Fig. 10(b) and 14(b). The plots of current drawn show that the faster is the detumbling rate the higher is the current drawn and the corresponding power consumption as shown in Figs. 8-14 (b). It can be noticed that the 2U CubeSat achieves steady state angular rates significantly faster than the 3U CubeSat for the same number of layers embedded. This clearly shows that as the dimensions of the spacecraft are increased to higher cube units, higher torque is demanded by the spacecraft for damping the body rates; therefore, embedded magnetorquer layers inside the PCB could be increased for achieving faster detumbling rates. Higher number of layers ($n \times m$) could be embedded inside the PCB for maximizing torque generation and consequently attaining the steady state proficiently if faster detumbling rates are prioritized. The steady state angular rate is then almost $\vec{\omega}_{B/I} = [0, 0, 0]^T$ rad/s for the possible configurations of asymmetric magnetorquers. The magnetorquer designs based on symmetrical planar windings do not detumble the CubeSat for the similar initial angular velocity conditions as shown in Fig. 11. This means that the conventional symmetric embedded magnetorquers cannot detumble the CubeSat at high angular rates and could only be used for attitude maneuverability. As the superiority of proposed coils with better detumbling rates has been established, steady state thermal analysis for single axis attitude maneuvers is required to compare the torque to power dissipation ratios with the embedded designs in available published literature.[1]

6. Thermal analysis

The thermal analyses are done using the equations derived in (Ali et al., 2015) which compute the power consumption and its subsequent temperature rise of the copper traces in the PCB for a 2U CubeSat. 2U CubeSat is taken as a case study in this section for a fair comparison of the torque to power dissipation ratio of asymmetric magnetorquers' with the embedded designs in published literature. The equations are as following:

$$T_o = \sqrt[4]{\frac{P_d + \alpha\sigma T_I{}^4 S}{\alpha\sigma S}}, P_d = I^2 R \tag{15}$$

Whereas T_I is the surrounding temperature, P_d is the power dissipated inside the PCB, σ is the Stefan-Boltzmann constant, T_o is the surface temperature, and α is the emissivity value of solar panel module which is represented in (16).

$$\alpha = \frac{P_d}{\sigma S\left(T_o{}^4 - T_I{}^4\right)} \tag{16}$$

The voltages in the range of 3.3 V are applied and modeled after CubeSats' power distribution bus. The magnetic moment generated and the torque imparted increases with corresponding decrease of TWR values ($1 \rightarrow 0$). As the TWR values decrease, the number of turns decreases, trace width increases and the resistance drastically reduces which permits more current to pass through the coil and subsequently increases the overall temperature. In case of linearly increasing Coil $C2$, the imparted torque is 5.1μNm, dipole moment generation is 0.10Am^2, the current is 0.37A, PCB temperature is 38.5°C and power dissipated is 1.26 W. In linearly decreasing C3 Coil's case, the imparted torque is 4.9μNm, dipole moment generation is 0.09Am^2, the current is 0.36A, PCB temperature is 37.6°C and power dissipated is 1.18 W. Quadratic maximum C4 and minimum C5 at central turns show the same behavior. In case of quadratic maximum Coil C4, the imparted torque is 4.45μNm, dipole moment generation is 0.08Am^2, the current is 0.34A, PCB temperature is 34.14°C and power dissipated is 0.8 W.

Fig. 15 illustrates the thermal characteristics of the asymmetric coils. Series configurations' (4×1) power con-

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sumption is less than the single coil (1 × 1) resulting in lesser temperature but generating the same magnetic moment. In this way, for reducing the rise in temperature, multiple copper layers can be embedded inside the PCB. For instance, in a linearly increasing $C2$'s case, the temperature rise of single coil (1 × 1) is 38.5°C while the rise in temperature of series configuration (4 × 1) is 28.57°C.

The torque versus power dissipation ratio of series configuration (4 × 1) for all the layouts (C1, $C2$, C3, C4 and C5) is the maximum as their power consumption is very low. Similarly, the torque versus power dissipation ratio of the parallel configuration (1 × 4) for all layouts is the lowest because of the higher power consumption The parallel configuration (1 × 4) of linearly increasing $C2$ generates the highest torque of 20.4 μNm but gives rise to a higher temperature of 71.07°C at power dissipation of 5.04 W. This peak temperature is the steady state temperature when the magnetorquer is energized for a longer time. The temperatures of the magnetorquers increases exponentially at higher operating voltages so power buses with a 3.3 V input range should be designed such as those used on CubeSats. Furthermore, the configuration of coils (1 × 4, 2 × 2, 1 × 4) using the OBC can be changed at any time and the parallel configuration (1 × 4) can be used with hybrid configuration (2 × 2) sequentially in a controlled manner for detumbling with excessively high angular rates if needed for maintaining the thermals.

Table 2 tabulates the comparison amongst the embedded magnetorquers and state of the art. GomSpace Nano Power p110U is internal to PCB like the proposed asymmetric magnetorquer; nonetheless, they offer suboptimal torque versus power ratio. The proposed asymmetric planar coil $C2$ provides detumbling rates at better torque to power dissipation ratio as compared to the symmetric pla-

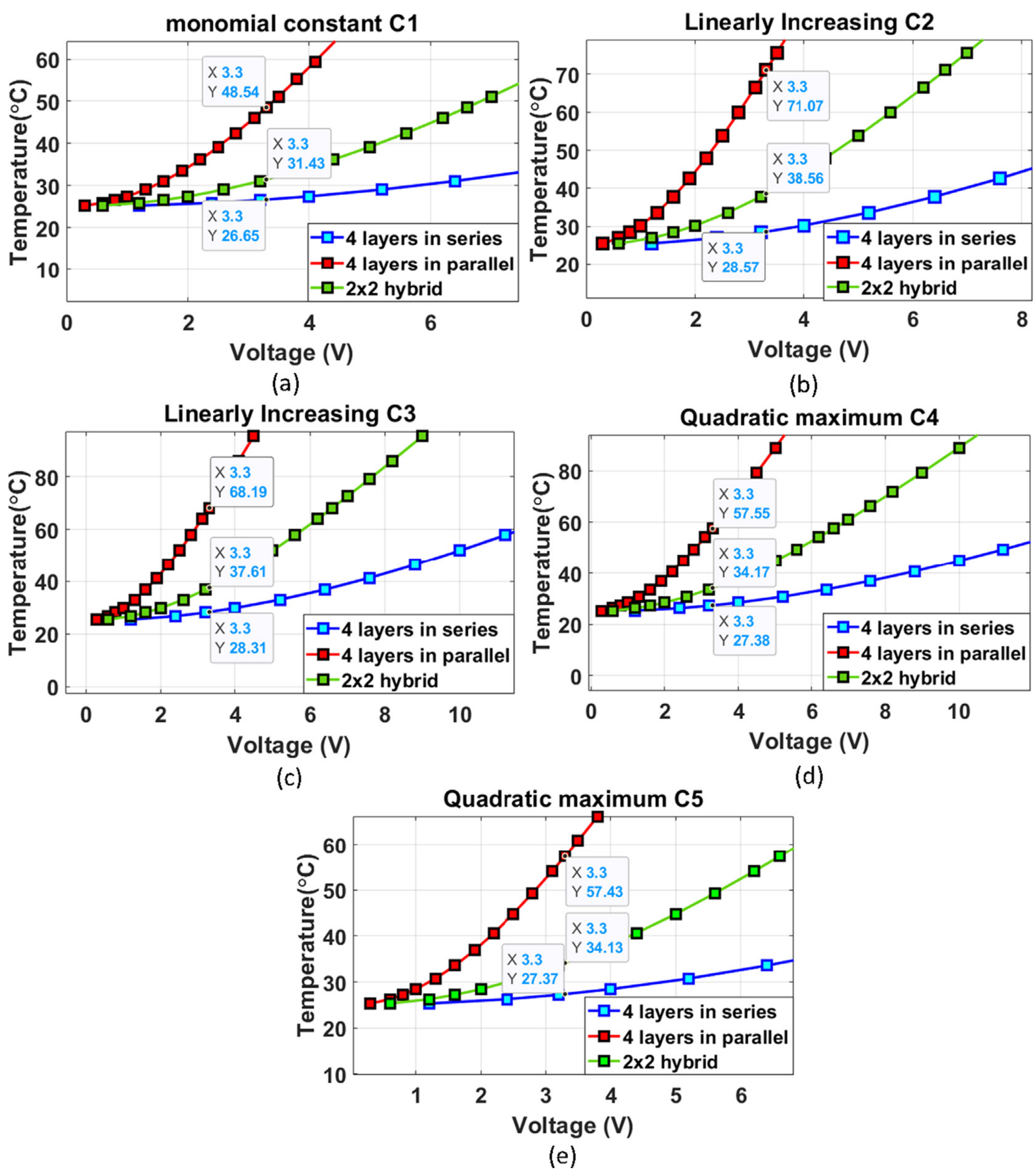

Fig. 15. Temperature rise of magnetorquer for various configurations against applied voltage.

Table 2
Torque to power dissipation ratio and rotation times for various configurations against applied voltage of 3.3 V.

Actuators		P (W)	τ(μNm)	$\frac{\tau}{P}$	Rotation time (s) $T = \sqrt{\frac{2I\varphi}{\tau}}$
Commercial design	Gomspace Nanopower (GomSpace, n.d.)	2.3	1.9	0.82	150.1
Embedded Printed magnetorquers	Series configuration (4 × 1) of ref (Ali et al., 2015)	1.26	2.45	1.94	86.8
	Series configuration (4 × 1) of ref (Mughal et al., 2020) (optimized design)	0.14	1.42	10.14	138.4
	Series configuration (4 × 1) of (proposed *C*2)	0.31	5.1	16.35	60.3
	Parallel configuration (1 × 4) of (proposed *C*2)	5.04	20.4	4.04	30.1
	Hybrid configuration (2 × 2) Coils (proposed C4)	0.84	8.84	10.5	45.7

nar coils in the published literature. The series (4 × 1) and hybrid (2 × 2) configurations of linearly increasing *C*2 coil feature optimal torque to power dissipation ratios in comparison with the best design symmetric cases of (Ali et al., 2015) and (Mughal et al., 2020). The rotation times given in Table 2 are for single axis attitude maneuvers along the arbitrary axis for 90° rotation. In conclusion, missions that need the spacecraft to maneuver at high precision should select the linearly increasing coil *C*2 while missions for a longer lifespan could opt for the coils based on quadratic function.

7. Conclusion

Asymmetric magnetorquers have been used to damp out the body rates of a tumbling spacecraft in low Earth Orbit. The proposed coils with non-unity track width ratio were used in higher form factor CubeSats at an altitude of 400 km. The results guarantee three-axis stability using three embedded magnetorquers without relying on bulky actuators while recovering from tumbles with high initial angular rates. It was found that conventional symmetric embedded magnetorquers in the literature for similar initial conditions could not detumble the spacecraft while all the proposed asymmetric magnetorquers reduced the initial angular velocities to almost zero. Due to the embedded design, thermal analyses of various coils have been performed to preserve thermal limits and show effectiveness of the system. The torque to power dissipation ratios given by the asymmetric magnetorquers are better than the embedded designs in the published literature and commercial state-of-the-art. Future work could be done on the proposed coils for the problem of pure magnetic fast detumbling for multi-cube higher form factor satellites with higher interia.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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